Physics 9702 Paper 4 · Topic 15

Ideal gases

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/41 October/November 2021, Question 3 (13 marks) page 2

Examiner report: A few candidates confused the symbols and treated n and N as the same quantity. (c) (i) Very few candidates stated the first law of thermodynamics in words although the symbol equation was often given.

2 9702/41 May/June 2021, Question 2 (10 marks) page 4

Examiner report: (b) This explanation proved to be challenging. (c) (i) Candidates found this calculation difficult.

3 9702/42 October/November 2022, Question 3 (10 marks) page 6

Examiner report: Common errors were confusion between number of molecules and amount of substance for N, and not being clear that T is thermodynamic temperature. Some of the weaker candidates were unable to make a start because they did not realise that the temperature needed to be converted from °C to K.

4 9702/42 October/November 2024, Question 4 (9 marks) page 8

Examiner report: A common misconception was that the assumption dealing with the negligible volume of the molecules related to a single molecule rather than to all of the molecules in the gas. Of the candidates that did address the question asked, most observed that there are molecular collisions with the walls of the container, but only the strongest candidates discussed the application of Newton's laws to those collisions to explain the origin of the force on the walls.

5 9702/42 February/March 2023, Question 2 (12 marks) page 10

Examiner report: (a) There was much confusion here between the definition of an ideal gas and the assumptions of the kinetic theory. A few candidates omitted the conversion from temperature in degrees Celsius to kelvin.

6 9702/42 February/March 2021, Question 2 (6 marks) page 12

Examiner report: There was much confusion between the mass of the gas and the mass of a molecule, and also between r.m.s. speed and the mean square speed.

7 9702/42 May/June 2021, Question 2 (9 marks) page 14

Examiner report: (ii) Many candidates found this question difficult, and often only got as far as calculating the change in kinetic energy of a single molecule.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

3 **(a)** Define *specific heat capacity*.

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..... [2]

(b) A sealed container of fixed volume V contains N molecules, each of mass m, of an ideal gas at pressure p.

(i) State an expression, in terms of V, N, p and the Boltzmann constant k, for the thermodynamic temperature T of the gas.

..... [1]

(ii) Show that the mean translational kinetic energy E_K of a molecule of the gas is given by

$$E_K = \frac{3}{2}kT.$$

[2]

(iii) Explain why the internal energy of the gas is equal to the total kinetic energy of the molecules.

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.....

..... [2]

(c) The gas in **(b)** is supplied with thermal energy Q.

(i) Explain, with reference to the first law of thermodynamics, why the increase in internal energy of the gas is Q.

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..... [2]

(ii) Use the expression in **(b)(ii)** and the information in **(c)(i)** to show that the specific heat capacity c of the gas is given by

$$c = \frac{3k}{2m}.$$

[2]

(d) The container in **(b)** is now replaced with one that does not have a fixed volume. Instead, the gas is able to expand, so that the pressure of the gas remains constant as thermal energy is supplied.

Suggest, with a reason, how the specific heat capacity of the gas would now compare with the value in **(c)(ii)**.

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..... [2]

[Total: 13]

2 An ideal gas is contained in a cylinder by means of a movable frictionless piston, as illustrated in Fig. 2.1.

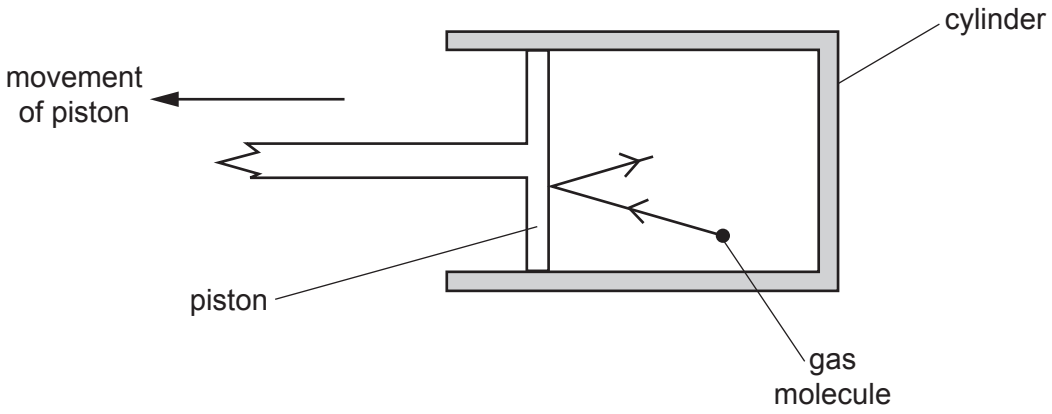


Fig. 2.1

Initially, the gas has a volume of $1.8 \times 10^{-3}\,m^3$ at a pressure of $3.3 \times 10^5\,Pa$ and a temperature of 310 K.

(a) Show that the number of gas molecules in the cylinder is 1.4×10^{23}.

[2]

(b) Use kinetic theory to explain why, when the piston is moved so that the gas expands, this causes a decrease in the temperature of the gas.

.....

.....

.....

..... [3]

(c) The gas expands so that its volume increases to $2.4 \times 10^{-3}\,m^3$ at a pressure of $2.3 \times 10^5\,Pa$ and a temperature of 288 K, as shown in Fig. 2.2.

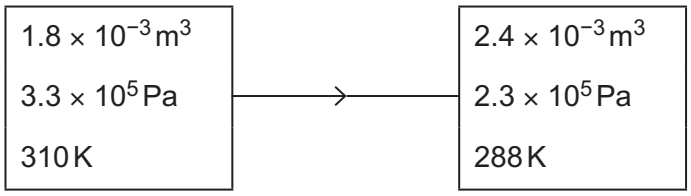


Fig. 2.2

(i) The average translational kinetic energy E_K of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT$$

where k is the Boltzmann constant and T is the thermodynamic temperature.

Calculate the increase in internal energy ΔU of the gas during the expansion.

ΔU = ... J [3]

(ii) The work done by the gas during the expansion is 76 J.

Use your answer in **(i)** to explain whether thermal energy is transferred to or from the gas during the expansion.

.....

.....

..... [2]

[Total: 10]

3 **(a)** The equation of state for an ideal gas can be written as

$$pV = NkT.$$

State the meaning of each of the symbols in this equation.

p: .....

V: .....

N: .....

k: .....

T: .....

[3]

(b) Use the equation in **(a)** to show that the average translational kinetic energy E_K of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT.$$

[2]

(c) The mass of an oxygen molecule is 5.31×10^{-26} kg. Assume that oxygen behaves as an ideal gas.

(i) Use the equation in **(b)** to determine the root-mean-square (r.m.s.) speed *u* of an oxygen molecule at 23 °C.

u = ... $m\,s^{-1}$ [3]

(ii) A fixed mass of oxygen gas at initial pressure *P* is sealed in a cylindrical container by a movable piston at one end, as shown in Fig. 3.1.

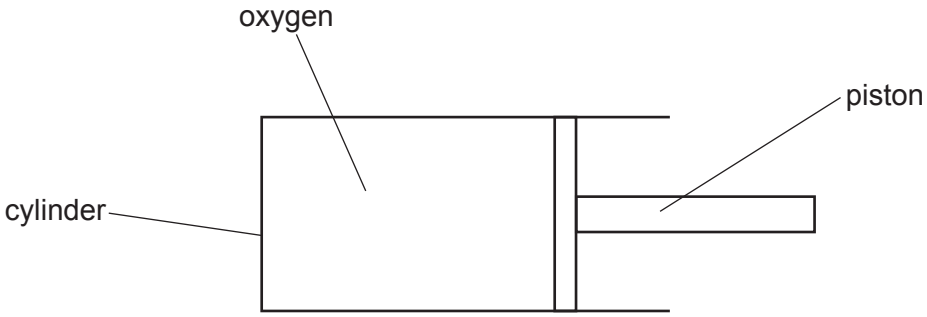


Fig. 3.1

The temperature of the gas is 23 °C.

The piston is slowly moved into the cylinder so that the oxygen gas is compressed. At all times, the gas and the container remain in thermal equilibrium with the surroundings.

On Fig. 3.2, sketch the variation with pressure of the r.m.s. speed of the oxygen molecules as the pressure increases.



Fig. 3.2

[2]

[Total: 10]

4 **(a)** State **three** of the basic assumptions of the kinetic theory of gases.

1 .....

.....

2 .....

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3 .....

.....

[3]

(b) Explain how molecular movement causes the pressure exerted by a gas.

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..... [3]

(c) Fig. 4.1 shows the variation with thermodynamic temperature T of the mean-square speeds $\langle c^2 \rangle$ for two gases X and Y.

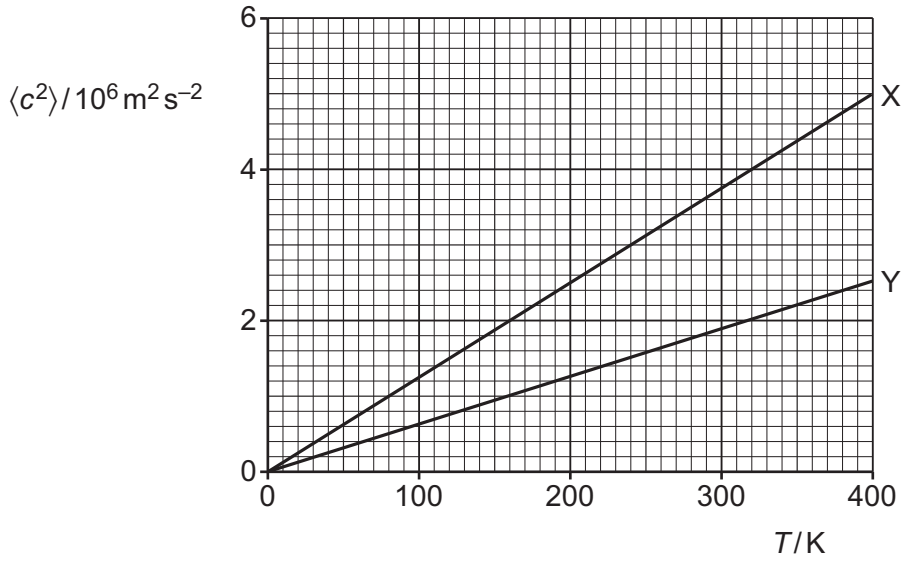


Fig. 4.1

Fig. 4.2 shows the variation with T of the product pV for samples of the two gases, where p is the pressure of the gas and V is the volume of the gas.

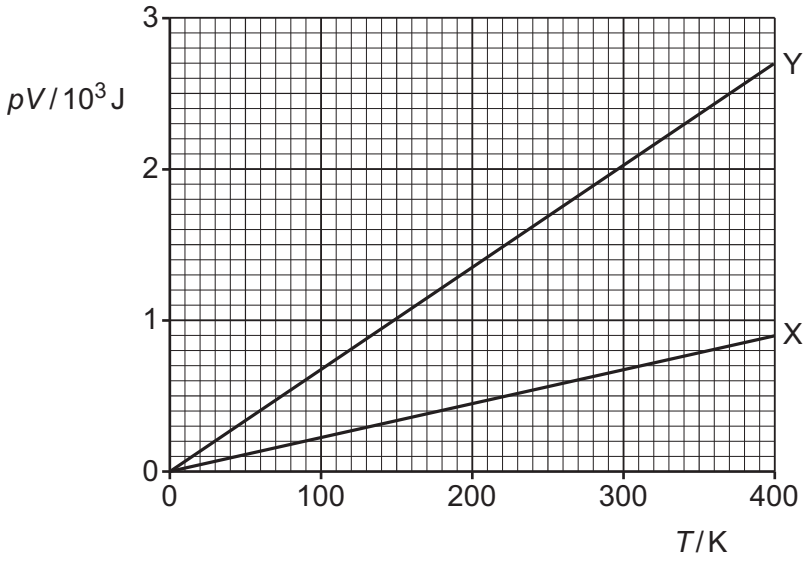


Fig. 4.2

State **three** conclusions about the gases and their samples that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working that you need.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 9]

2 **(a)** State what is meant by an ideal gas.

.....

.....

..... [2]

(b) A fixed amount of helium gas is sealed in a container. The helium gas has a pressure of 1.10×10^5 Pa, and a volume of 540 cm^3 at a temperature of 27 °C.

The volume of the container is rapidly decreased to 30.0 cm^3. The pressure of the helium gas increases to 6.70×10^6 Pa and its temperature increases to 742 °C, as illustrated in Fig. 2.1.

initial state	final state
1.10×10^5 Pa	6.70×10^6 Pa
540 cm^3	30.0 cm^3
27 °C	742 °C

Fig. 2.1

No thermal energy enters or leaves the helium gas during this process.

(i) Show that the helium gas behaves as an ideal gas.

[2]

(ii) The first law of thermodynamics may be expressed as

$$\Delta U = q + W.$$

Use the first law of thermodynamics to explain why the temperature of the helium gas increases.

.....

.....

.....

.....

..... [2]

(iii) The average translational kinetic energy E_K of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT$$

where k is the Boltzmann constant and T is the thermodynamic temperature.

Calculate the change in the total kinetic energy of the molecules of the helium gas.

change in kinetic energy = ... J [3]

(c) The mass of nitrogen gas in another container is 24.0 g at a temperature of 27 °C. The gas is cooled to its boiling point of –196 °C. Assume all the gas condenses to a liquid.

For this change the specific heat capacity of nitrogen gas is 1.04 $kJ\,kg^{-1}\,K^{-1}$.

The specific latent heat of vaporisation of nitrogen is 199 $kJ\,kg^{-1}$.

Determine the thermal energy, in kJ, removed from the nitrogen gas.

energy = ... kJ [3]

[Total: 12]

2 A fixed mass of an ideal gas is at a temperature of 21 °C. The pressure of the gas is 2.3×10^5 Pa and its volume is $3.5 \times 10^{-3}\,m^3$.

(a) (i) Calculate the number N of molecules in the gas.

N = ... [2]

(ii) The mass of one molecule of the gas is 40 u.

Determine the root-mean-square (r.m.s.) speed of the gas molecules.

r.m.s. speed = ... $m\,s^{-1}$ [2]

(b) The temperature of the gas is increased by 84 °C.

Calculate the value of the ratio

$$\frac{\text{new r.m.s. speed of molecules}}{\text{original r.m.s. speed of molecules}}.$$

ratio = ... [2]

[Total: 6]

2 An ideal gas has a volume of $3.1 \times 10^{-3}\,m^3$ at a pressure of $8.5 \times 10^5\,Pa$ and a temperature of 290 K, as shown in Fig. 2.1.

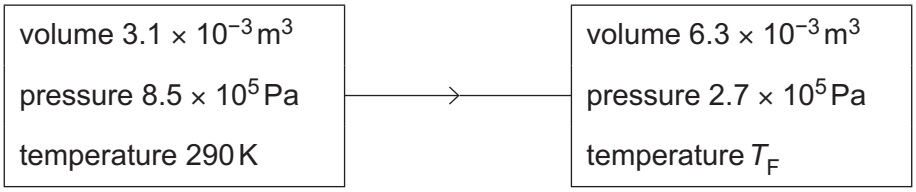


Fig. 2.1

The gas suddenly expands to a volume of $6.3 \times 10^{-3}\,m^3$. During the expansion, no thermal energy is transferred. The final pressure of the gas is $2.7 \times 10^5\,Pa$ at temperature T_F, as shown in Fig. 2.1.

(a) Show that the number of gas molecules is 6.6×10^{23}.

[3]

(b) (i) Show that the final temperature T_F of the gas is 190 K.

[1]

(ii) The average translational kinetic energy E_K of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT$$

where T is the thermodynamic temperature and k is the Boltzmann constant.

Calculate the increase in internal energy ΔU of the gas.

ΔU = ... J [3]

(c) Use the first law of thermodynamics to explain why the external work w done on the gas during the expansion is equal to the increase in internal energy in **(b)(ii)**.

.....

.....

..... [2]

[Total: 9]

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause temperature change)	**B1**
	(thermal) energy per unit change in temperature	**B1**
3(b)(i)	(T =) pV / Nk	**B1**
3(b)(ii)	(pV =) $NkT = \frac{1}{3}Nm\langle c^2\rangle$ **or** $pV = NkT$ **and** $pV = \frac{1}{3}Nm\langle c^2\rangle$	**M1**
	leading to $\frac{1}{2}m\langle c^2\rangle = (3/2)kT$ and $\frac{1}{2}m\langle c^2\rangle = E_K$	**A1**
3(b)(iii)	internal energy = ΣE_K (of molecules) + ΣE_P (of molecules) **or** no forces between molecules	**B1**
	potential energy of molecules is zero	**B1**
3(c)(i)	increase in internal energy = Q + work done	**B1**
	constant volume so no work done	**B1**
3(c)(ii)	$c = Q / Nm\Delta T$	**C1**
	$= [N \times (3/2)k\Delta T] / (Nm\Delta T) = 3k / 2m$	**A1**
3(d)	(as it expands) gas does work (against the atmosphere/external pressure)	**B1**
	for same temperature rise) more (thermal) energy needed, so larger specific heat capacity	**B1**

Question	Answer	Marks
2(a)	$pV = NkT$	**C1**
	$N = (1.8 \times 10^{-3} \times 3.3 \times 10^{5}) / (1.38 \times 10^{-23} \times 310) = 1.4 \times 10^{23}$	**A1**
	or	
	$pV = nRT$ **and** $nN_A = N$	**(C1)**
	$N = (1.8 \times 10^{-3} \times 3.3 \times 10^{5} \times 6.02 \times 10^{23}) / (8.31 \times 310) = 1.4 \times 10^{23}$	**(A1)**
2(b)	speed of molecule decreases on impact with moving piston	**B1**
	mean square speed (directly) proportional to (thermodynamic) temperature **or** mean square speed (directly) proportional to kinetic energy (of molecules) **or** kinetic energy (of molecules) (directly) proportional to (thermodynamic) temperature	**B1**
	kinetic energy (of molecules) decreases (so temperature decreases)	**B1**
2(c)(i)	$\Delta U = 3/2 \times k \times \Delta T \times N$	**C1**
	$= 3/2 \times 1.38 \times 10^{-23} \times (288 - 310) \times 1.4 \times 10^{23}$	**C1**
	$= -64$ J	**A1**
2(c)(ii)	decrease in internal energy is less than work done by gas	**M1**
	(thermal energy is) transferred <u>to</u> the gas (during the expansion)	**A1**

Question	Answer	Marks
3(a)	p = pressure (of gas), V = volume (of gas) **and** k = Boltzmann constant	**B1**
	N = number of molecules	**B1**
	T = thermodynamic temperature	**B1**
3(b)	($pV = NkT$ and $pV = \frac{1}{3}Nm\langle c^2\rangle$ leading to) $NkT = \frac{1}{3}Nm\langle c^2\rangle$	**M1**
	algebra leading to $(3/2)kT = \frac{1}{2}m\langle c^2\rangle$ **and** use of $\frac{1}{2}m\langle c^2\rangle = E_K$ leading to $(3/2)kT = E_K$	**A1**
3(c)(i)	$T = 296$ K	**C1**
	$\frac{1}{2}m\langle c^2\rangle = (3/2)kT$ $\frac{1}{2} \times 5.31 \times 10^{-26} \times u^2 = (3/2) \times 1.38 \times 10^{-23} \times 296$	**C1**
	$u = 480\ \text{m s}^{-1}$	**A1**
3(c)(ii)	line passing through (P, u)	**B1**
	horizontal straight line	**B1**

Question	Answer	Marks
4(a)	• molecules are in (constant) random motion • (all) collisions between molecules are (perfectly) elastic • no forces between molecules (except during collisions) • volume of molecules is negligible (compared with volume of gas) • collisions involving molecules are instantaneous *Any three points, 1 mark each*	**B3**
4(b)	• molecules collide with (walls of) container • momentum of molecule changes during collision (with walls) • change in momentum is caused by force on molecule by wall • molecule experiences force from wall so molecule exerts force on wall • many molecules exerting force across the area of the wall leads to pressure (on the wall) *Any three points, 1 mark each*	**B3**
4(c)	*Any three bulleted points from:* • both gases are ideal *Up to 2 points from:* • mass of one molecule of gas X is 3.3×10^{-27} kg • mass of one molecule of gas Y is 6.6×10^{-27} kg • mass of one molecule of gas Y is double mass of one molecule of gas X *Up to 2 points from:* • sample of X contains 0.27 mol / 1.6×10^{23} molecules • sample of Y contains 0.81 mol / 4.9×10^{23} molecules • sample of Y contains treble the amount of gas / number of molecules as sample of X *Up to 2 points from:* • mass of gas X is 5.4×10^{-4} kg • mass of gas Y is 3.2×10^{-3} kg • mass of gas Y is six times mass of gas X	**B3**

Question	Answer	Marks
2(a)	gas for which $pV \propto T$	**M1**
	where T is thermodynamic temperature	**A1**
2(b)(i)	evidence of two temperature conversions between °C and K	**B1**
	two calculations shown, one for each state e.g. $\frac{1.10\times10^{5}\times540\times10^{-6}}{(273+27)} = 0.198$ **and** $\frac{6.70\times10^{6}\times30\times10^{-6}}{(273+742)} = 0.198$	**A1**
2(b)(ii)	work is done on the gas	**M1**
	internal energy increases (so temperature increases)	**A1**

Question	Answer	Marks
2(b)(iii)	$pV = NkT$ e.g. $N = \frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$ $= 1.435 \times 10^{22}$ $\Delta E_k = (3/2)\, k \Delta T N$	**C1**
	$= (3/2) \times 1.38 \times 10^{23} \times (742 - 27) \times \frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$	**C1**
	$= 212$ J	**A1**
2(c)	$E = mc\Delta\theta$ **and** $E = mL$	**C1**
	$\Delta\theta$ = (27 + 196) or 223	**C1**
	$E = 0.0240 \times 1.04 \times (27 + 196) + 0.0240 \times 199$ $= 10.3$ kJ	**A1**

Question	Answer	Marks
2(a)(i)	$pV = NkT$ or $pV = nRT$ and $N = nN_A$ $N = \frac{2.3\times10^{5}\times3.5\times10^{-3}}{1.38\times10^{-23}\times294}$	**C1**
	= **2.0×10^{23}**	**A1**
2(a)(ii)	$pV = \frac{1}{3}Nmc^2$ $c^2 = \frac{3\times2.3\times10^{5}\times3.5\times10^{-3}}{2.0\times10^{23}\times40\times1.66\times10^{-27}}$ = 182 000 r.m.s. speed = **430** m s^{-1}	**C1**
	or $\frac{1}{2}mc^2 = \frac{3}{2}kT$	**A1**
	$c^2 = \frac{3\times1.38\times10^{-23}\times294}{40\times1.66\times10^{-27}}$ = 183 000	**(C1)**
	r.m.s.speed = 430 m s^{-1}	**(A1)**

Question	Answer	Marks
2(b)	$c^2 = \frac{3\times 2.0\times 10^{23}\times 1.38\times 10^{-23}\times(294+84)}{2.0\times 10^{23}\times 40\times 1.66\times 10^{-27}}$ $c^2 = 236\,000$ $c = 485$	**C1**
	$ratio\left(=\frac{485}{430}\right) = 1.1$	**A1**
	OR $v \propto \sqrt{T}$ or $v^2 \propto T$	**(C1)**
	$ratio = \sqrt{\frac{273+21+84}{273+21}}$ or $\sqrt{\frac{378}{294}}$ ratio = **1.1**	**(A1)**

Question	Answer	Marks
2(a)	$pV = nRT$	**C1**
	$pV = nRT$ **and** $N = nN_A$ **or** $pV = NkT$	**C1**
	$3.1 \times 10^{-3} \times 8.5 \times 10^5 = (N \times 290 \times 8.31) / (6.02 \times 10^{23})$ so $N = 6.6 \times 10^{23}$ **or** $3.1 \times 10^{-3} \times 8.5 \times 10^5 = N \times 1.38 \times 10^{-23} \times 290$ so $N = 6.6 \times 10^{23}$	**A1**
2(b)(i)	$(3.1 \times 10^{-3} \times 8.5 \times 10^5) / 290 = (6.3 \times 10^{-3} \times 2.7 \times 10^5) / T$ so $T = 190$ K **or** $6.3 \times 10^{-3} \times 2.7 \times 10^5 = 6.6 \times 10^{23} \times 1.38 \times 10^{-23} \times T$ so $T = 190$ K	**A1**
2(b)(ii)	$\Delta U = 3/2 \times k \times \Delta T \times N$	**C1**
	$= 3/2 \times 1.38 \times 10^{-23} \times (190 - 290) \times 6.6 \times 10^{23}$	**C1**
	$= -1400$ J	**A1**
2(c)	$\Delta U = q + w$	**M1**
	$q = 0$ so $\Delta U = w$	**A1**